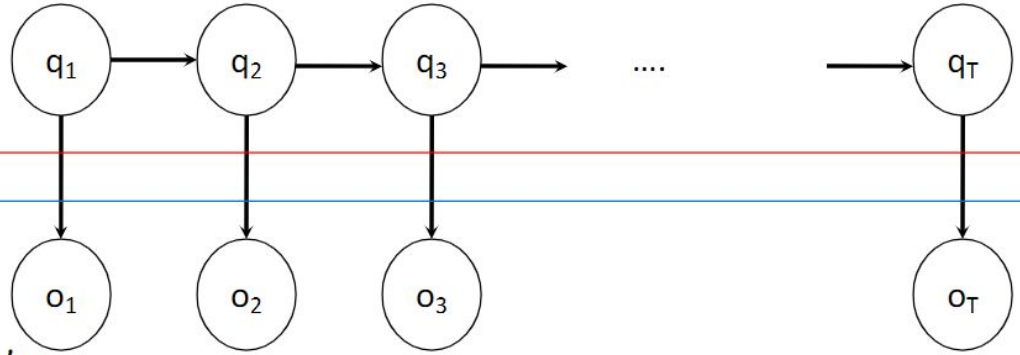
Recitation 3/28

Hidden Markov Models

Hidden Markov Models

q_i's:

- Random variables that take values from a set S = {s_1 … s_n}, e.g., {I,E,B} in the gene example
- S is called "states" in HMM literature



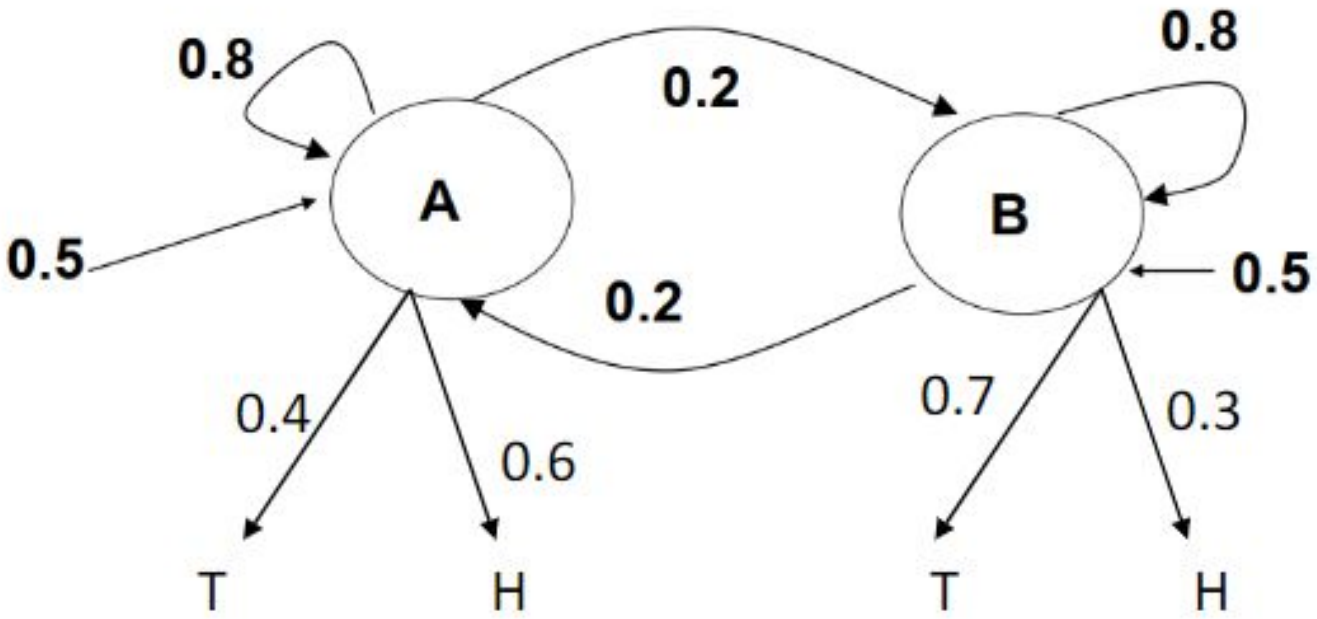
o_i's:

- Random variables that take values from a set Σ={σ_1, …, σ_m}, e.g., {A,C,T,G} in the gene example
- Called "outputs" in HMM literature

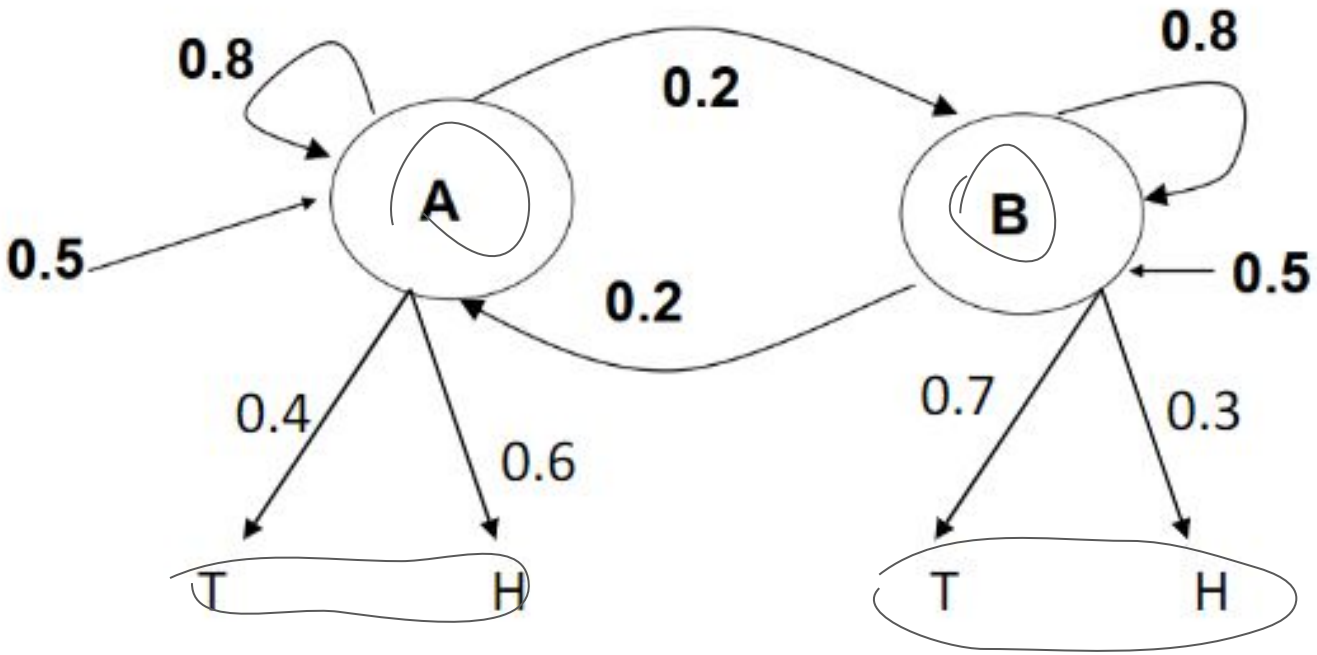
HMM Definitions

- **States:** Classes the sequence currently is in. For instance, exon, promoter, etc.
- **Initial State Probability:** Probabilities for each possible state at q1. (S)
- **Transition Probability:** Probability of going to one state from current state. (S x S)
- **Emission Probability:** Probability of an observation, given the current state. (S x O)

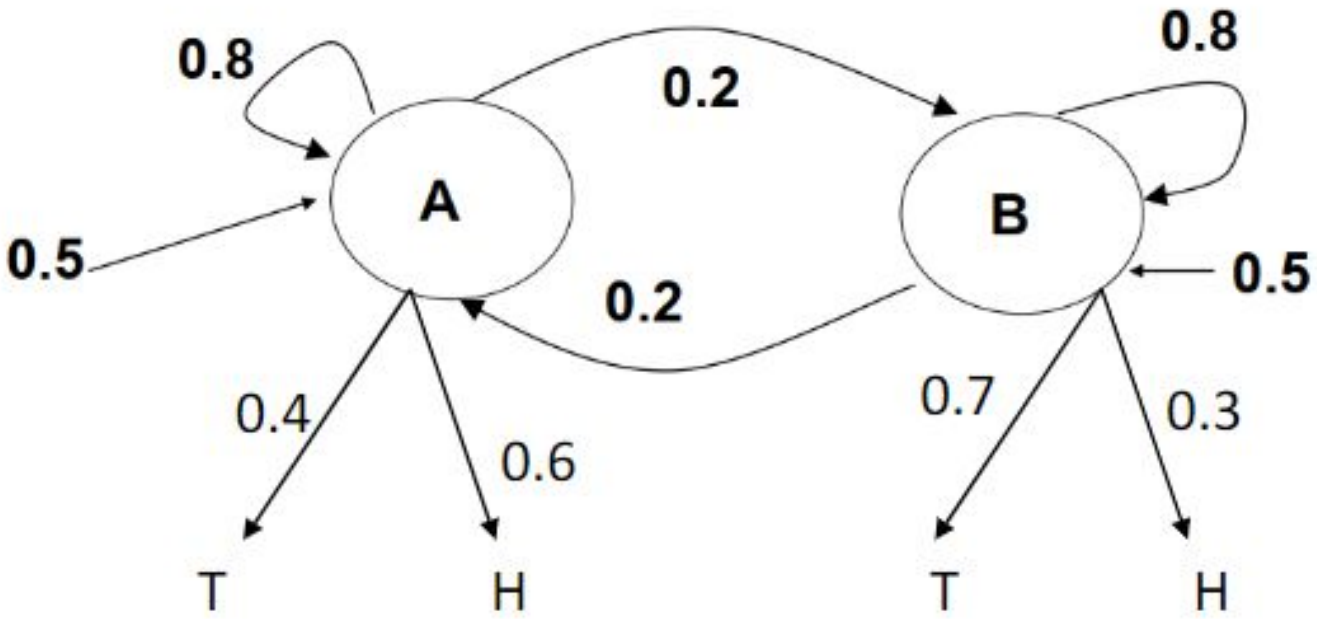
Identify States and Observations



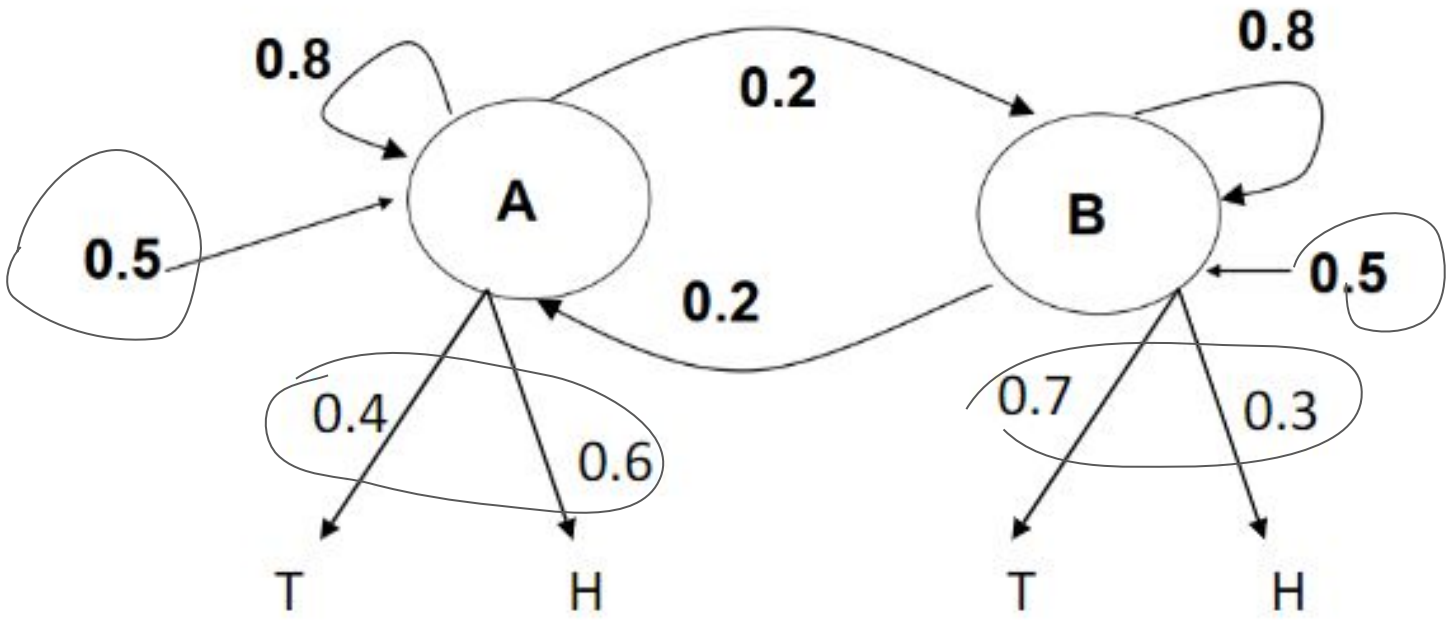
Identify States and Observations



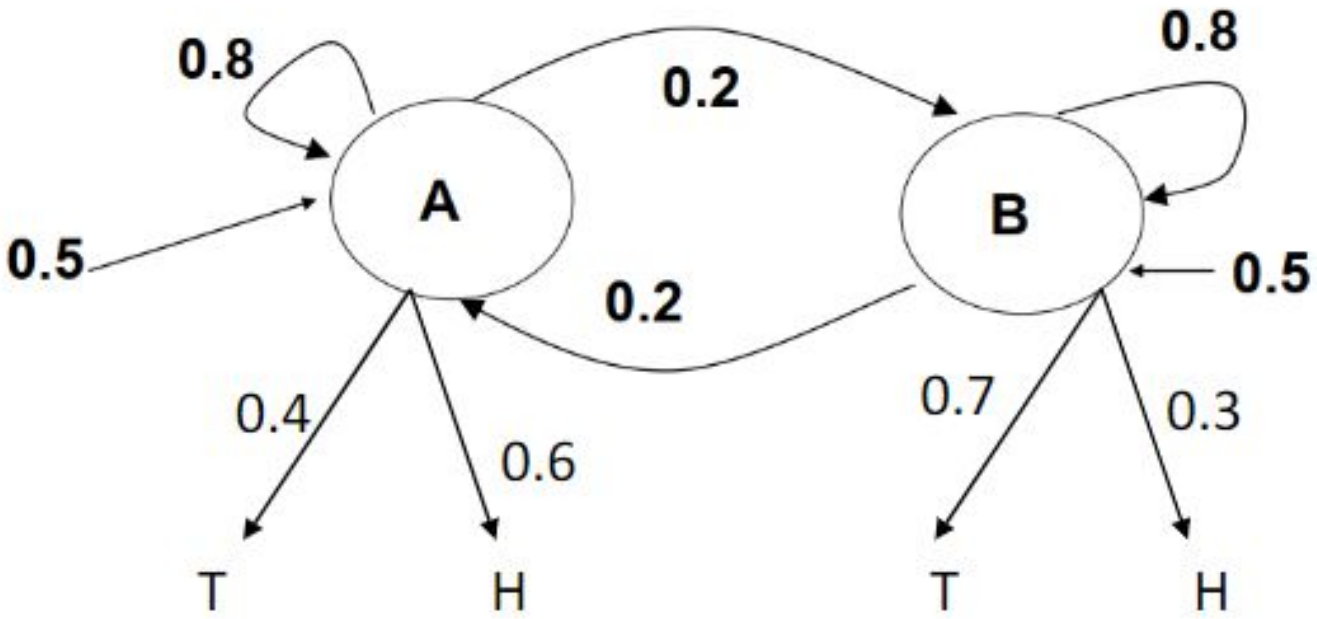
Identify Initial Probabilities and Emission Probabilities



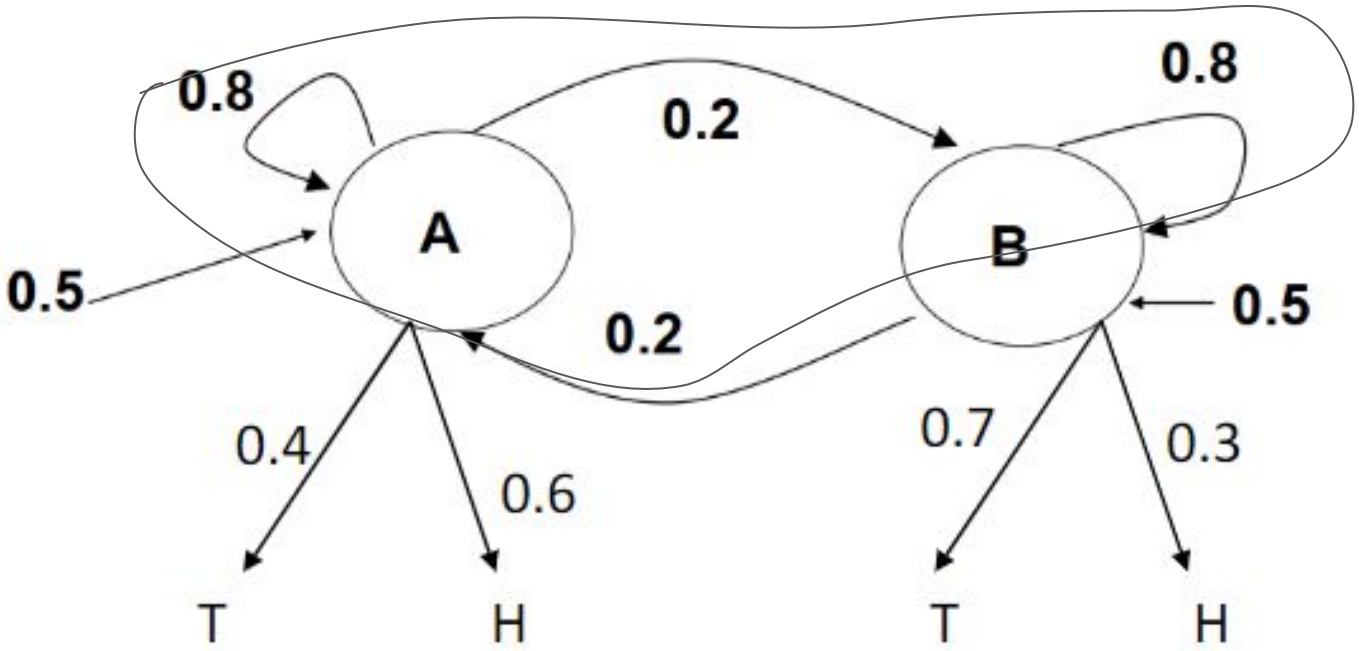
Identify Initial Probabilities and Emission Probabilities

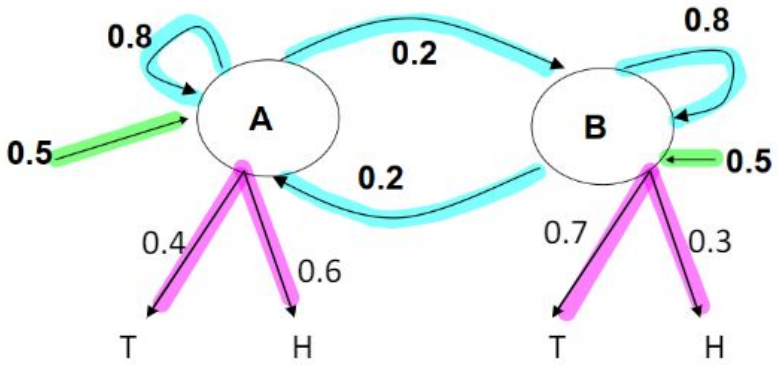


Identify Transition Probabilities



Identify Transition Probabilities





These are the parameters!

Initial Probabilities:

$P(Q1 = A) = 0.5$
$P(Q1 = B) = 0.5$

Transition Probabilities:

$P(Q_{t+1} = A \mid Q_t = A) = 0.8$
$P(Q_{t+1} = B \mid Q_t = A) = 0.2$
$P(Q_{t+1} = A \mid Q_t = B) = 0.2$
$P(Q_{t+1} = B \mid Q_t = B) = 0.8$

Emission Probabilities:

$P(O_t = T \mid Q_t = A) = 0.4$
$P(O_t = H \mid Q_t = A) = 0.6$
$P(O_t = T \mid Q_t = B) = 0.7$
$P(Q_t = H \mid Q_t = B) = 0.3$

Calculating P(States, Observations)

Identify the starting probability, emission probability, and transition probabilities in here.

b_q(o) means probability of observation o from state q.

$$P(Q, O) = \pi_{q_1} \cdot b_{q_1}(o_1) \cdot \prod_{t=2}^{T} a_{q_{t-1} q_t} \cdot b_{q_t}(o_t)$$

Forward Backward Algorithm

- Essential idea:
  - Compute forward pass to give \alpha (states by observations)
  - Compute backward pass to give \beta (states by observations)
- Alpha and Beta represent probabilities for each state at each observation.
- Forward pass: “Given the HMM parameters, what is the probability of seeing the observed sequence up to this point and ending in a specific state?”
- Backward pass: “Starting in a specific state, how likely is it to observe the rest of the sequence?”

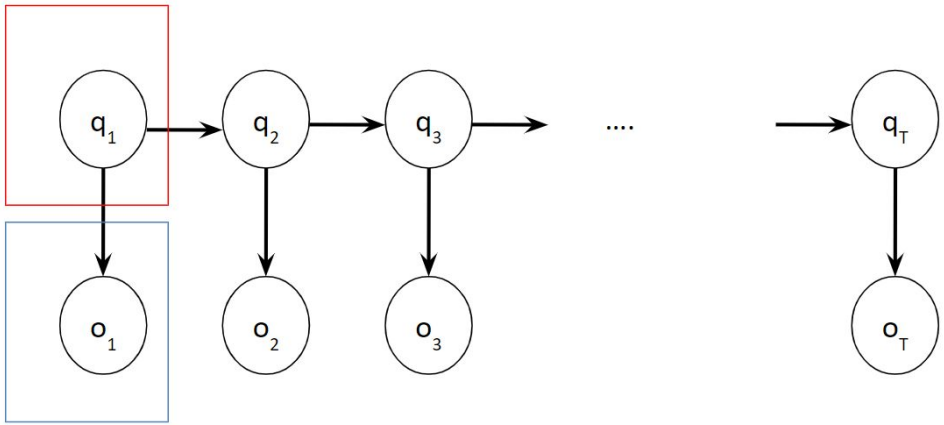
Forward algorithm - Initialization

Goal: $\alpha_t(i) = P(o_1, \cdots, o_t, q_t = s_i)$ for $t = 1, \cdots, T$ and $s_i = 1, \cdots, n$

- Initialization: what's $\alpha_1(i)$?

$\alpha_1(i) = P(o_1, q_1 = s_i)$ Definition

$= P(o_1 | q_1 = s_i) P(q_1 = s_i)$ Emission + initial state

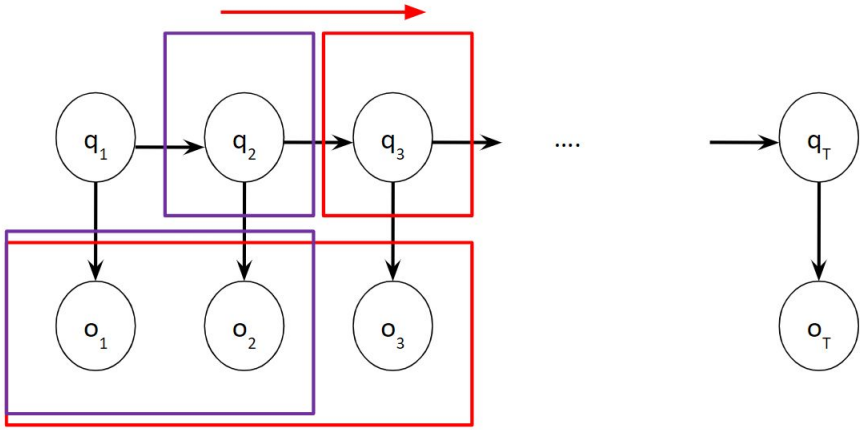


Forward Algorithm - Iteration

Goal: $\alpha_t(i) = P(o_1, \cdots, o_t, q_t = s_i)$ for $t = 1, \cdots, T$ and $s_i = 1, \cdots, n$

- Forward recursion: update $\alpha_{t+1}(i)$ given $\alpha_t(j), j = 1, \cdots, n$?

$$\alpha_{t+1}(i) = \Sigma_{j=1}^{n} P(o_{t+1}|q_{t+1} = s_i) P(q_{t+1} = s_i|q_t = s_j) \alpha_t(j)$$



Probability of observations

- Ex. P(ATGACTACA)

$$P(O) = \sum_i P(o_1, \cdots, o_T, q_T = s_i)$$

Probability of observations

- Ex. P(ATGACTACA)

$$P(O) = \sum_i P(o_1, \cdots, o_T, q_T = s_i)$$

- Alternatively, $\sum_i \alpha_T(i)$

Probability of States Given Observations

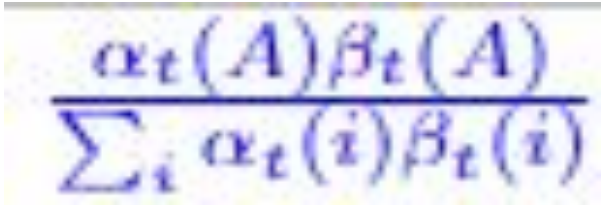
- P(EEEPPH|ATACAG)
- P(Q|O) = ?

Probability of States Given Observations

- P(EEEPPH|ATACAG)
- P(Q|O) = **P(Q, O) / P(O)**

Probability of state at time step given observations

- P(q_t = A|O)
- Alpha represents probability of observing previous observations and ending up in state.
- Beta represents probability of observing following observations given this state.



Compute Most Probable Path

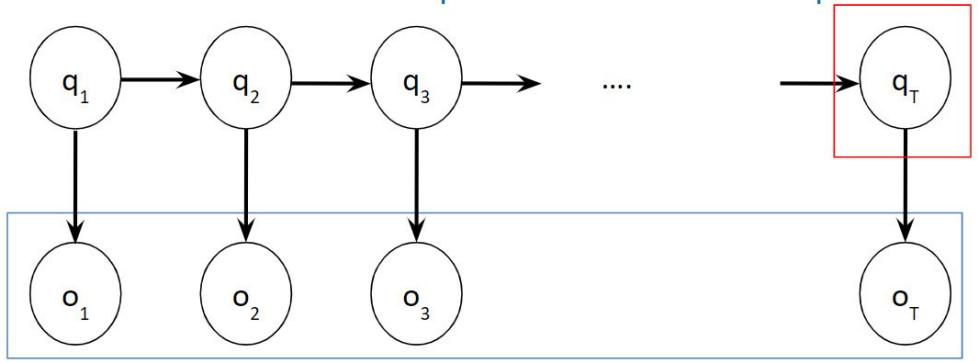
$$\arg\max_Q P(Q \mid O) = \arg\max_Q \frac{P(O \mid Q)P(Q)}{P(O)}$$

$$= \arg\max_Q P(O \mid Q)P(Q)$$

Viterbi Algorithm

- $\delta_t(i)$: maximum probability of ending up at state s_i at observation t (out of all possible sequences of states leading up to it)
- Initialization: $\delta_1 = P(o_1|q_1 = s_i)P(q_1 = s_i)$
- Update $\delta_{t+1}(i) = \max_j P(o_{t+1}|q_{t+1} = s_i)P(q_{t+1} = s_i|q_t = s_j)\delta_t(j)$

Emission probabilities Transition probabilities



	t=1	t=2	t=3	...	t=T
s_1	$\delta_1(1)$	$\delta_2(1)$	$\delta_3(1)$		$\delta_T(1)$
...	...	...	...		...
s_n	$\delta_1(n)$	$\delta_2(n)$	$\delta_3(n)$		$\delta_T(n)$

Viterbi Questions

1. How many times do we iterate through the sequence?
2. What are the dimensions of the delta matrix?
3. What is stored at each delta value in the matrix?
4. Given a delta matrix, how do we find the best path?

Viterbi Questions

1. How many times do we iterate through the sequence?
    a. **Just one**
2. What are the dimensions of the delta matrix?
    a. **States x Observations**
3. Given a delta matrix, how do we find the best path?
    a. **Start at the last column, with the highest delta, move backwards, taking the highest delta each column.**
4. What is stored at each delta value in the matrix?
    a. **The probability of that state at that time point.**

Viterbi Algorithm Code

Initialization:

Viterbi Algorithm Code

Initialization:

1. First column of delta matrix becomes initial probability * emission probability given o and each state

Viterbi Algorithm Code

Initialization:

1. First column of delta matrix becomes initial probability * emission probability given o and each state

Iteration:

Viterbi Algorithm Code

Initialization:

1. First column of delta matrix becomes initial probability * emission probability given o and each state

Iteration:

1. Next column of delta matrix:

Viterbi Algorithm Code

Initialization:

1. First column of delta matrix becomes initial probability * emission probability given o and each state

Iteration:

1. Next column of delta matrix:
    a. For each state:

Viterbi Algorithm Code

Initialization:

1. First column of delta matrix becomes initial probability * emission probability given o and each state

Iteration:

1. Next column of delta matrix:
    a. For each state:
        i. Maximum of Emission * Transmission * Delta

Viterbi Algorithm Code

Initialization:

1. First column of delta matrix becomes initial probability * emission probability given o and each state

Iteration:

1. Next column of delta matrix:
    a. For each state:
        i. Maximum of Emission * Transmission * Delta
2. Repeat

Viterbi Algorithm Code

Initialization:

1. First column of delta matrix becomes initial probability * emission probability given o and each state

Iteration:

1. Next column of delta matrix:
    a. For each state:
        i. Maximum of Emission * Transmission * Delta
2. Repeat
3. When finished, find best path by starting at the end, and tracing back highest deltas in each column.

- $\delta_t(i)$: maximum probability of ending up at state s_i at observation t (out of all possible sequences of states leading up to it)
- Initialization: $\delta_1 = P(o_1|q_1 = s_i)P(q_1 = s_i)$
- Update $\delta_{t+1}(i) = \max_j P(o_{t+1}|q_{t+1} = s_i)P(q_{t+1} = s_i|q_t = s_j)\delta_t(j)$

Emission probabilities Transition probabilities

Sequence = AAC

Transition Probabilities:

	State 1	State 2	State 3
State 1	0.2	0.5	0.3
State 2	0.3	0.4	0.3
State 3	0.4	0.3	0.3

Initial State Probabilities:

State 1	State 2	State 3
0.4	0.3	0.3

Emission Probabilities:

	A	C	G	T
State 1	0.1	0.2	0.3	0.4
State 2	0.25	0.25	0.25	0.25
State 3	0.4	0.3	0.2	0.1